DEVOPS WITH MULTI CLOUD (AWS + AZURE) SYLLABUS

INTRODUCTION OF DEVOPS

- What is DevOps
- Evolution from traditional software development to DevOps
- Why organizations use DevOps
- Goals and benefits of DevOps
- DevOps Life Cycle
- How Cloud Relates to DevOps

INTRODUCTION OF CLOUD

- What is Cloud Computing
- Evolution from traditional computing to cloud
- Types of cloud (Service & Deployment Models)
- Cloud Providers
- Market value of cloud

AMAZON WEB SERVICES:

- ELASTIC COMPUTE CLOUD (EC2)
- VIRTUAL PRIVATE CLOUD (VPC)
- SIMPLE STORAGE SERVICE (S3)
- ELASTIC BLOCK STORAGE (EBS)
- IDENTITY AND ACCESS MANAGEMENT (IAM)
- RELATIONAL DATABASE SERVICE (RDS)
- ELASTIC LOAD BALANCER (ELB)
- AUTO SCALING GROUP (ASG)
- ELASTIC CONTAINER REGISTRY (ECR)
- ELASTIC CONTAINER SERVICE (ECS)
- ELASTIC KUBERNETES SERVICE (EKS)
- SIMPLE NOTIFICATION SERVICE (SNS)
- CLOUD WATCH
- ROUTE 53

AZURE SERVICES:

- Azure Active Directory
- Virtual Machines
- VM Scale Set
- Azure Load Balancers
- Azure Container Registry
- Azure Container Interface
- Azure Kubernetes Service
- Azure Pipelines
- Azure SQL Database
- Azure Blog Storage
- Azure Discs
- Azure Atrifacts
- Azure Key Valut

OS - LINUX:

- Hardware Information Commands
- System Information Commands
- File & Directory Commands
- File Viewing & Editing
- File Permissions & Ownership
- User & Group Management
- Process Management
- Service Management (Systemd)
- Networking Commands
- Package Management
- Archive & Compression

- Disk & Storage Management
- Log Management
- Text Processing (Very Important)

SCM/VCS - GIT, GITHUB & BITBUCKET

- Introduction to Version Control System (VCS)
- Introduction to Git
- Git Architecture & Concepts
- Basic Git Commands
- Branching and Merging
- Working with Remote Repositories
- Conflict Resolution
- Git Tags and Releases
- Introduction to GitHub
- Introduction to Bitbucket

CI/CD - JENKINS

1. Introduction to Jenkins
   - What is Jenkins?
   - Why Jenkins is used
   - Features of Jenkins
   - Benefits of automation
   - Jenkins in DevOps
2. Continuous Integration (CI) & Continuous Delivery (CD)
   - What is CI?
   - What is CD?
   - CI/CD pipeline overview
   - Manual vs automated deployment
3. Jenkins Architecture
   - Topics:
   - Jenkins Server (Controller)
   - Jenkins Agent (Worker/Node)
   - Build Executor
   - Jobs
   - Pipeline execution
4. Jenkins Installation and Setup
   - System requirements
   - Install Jenkins
   - Access Jenkins dashboard
   - Initial configuration
   - Unlock Jenkins
   - Install recommended plugins
5. Jenkins User Interface
   - Dashboard
   - Build Queue
   - Job Management
   - Console Output
   - Plugin Manager
   - System Configuration
6. Jenkins Jobs
   - Types of jobs:
   - Freestyle Project
   - Pipeline Project
   - Multi-branch Pipeline
   - Folder Organization
   - Hands-on:
   - Create job
   - Configure job
   - Run build

- View logs

7. Jenkins Pipeline
   - What is Pipeline?
   - Pipeline as Code
   - Pipeline stages
   - Declarative vs Scripted pipeline
8. Jenkinsfile
   - Introduction to Jenkinsfile
   - Writing pipeline code
   - Storing Jenkinsfile in Git
9. Integrating Jenkins with Git
   - Topics:
   - Connect Git repositories
   - Webhooks
   - Automatic builds
   - Branch builds
10. Build Automation
    - Build triggers
    - Poll SCM
    - Scheduled builds
    - Parameterized builds
11. Jenkins Plugins
    - Common plugins:
    - Git Plugin
    - Pipeline Plugin
    - Docker Plugin
    - Kubernetes Plugin
    - Sonarqube Plugin
    - Email Extension
12. Jenkins Security
    - User management
    - Authentication
    - Authorization
    - Credentials management
    - Secret handling
13. Monitoring and Troubleshooting
    - Build logs
    - Console output
    - Failed build analysis
    - Workspace cleanup

CONTAINERIZATION TOOL:

1. Introduction
   - Monolithic Vs Microservices
   - What is Docker
   - Life Before Docker
   - Docker Architecture
   - Installing and setup Docker
2. Containers
   - What is Containers
   - How to create Containers
   - Container Management (start, stop, kill, restart)
   - Container Life Cycle
3. Images
   - What is an image?
   - Pulling images from Docker Hub
   - Listing images
   - How to create custom Images
   - How to secure Images
   - How to reduce image sizes

4. Dockerfile
   - What is Dockerfile
   - Dockerfile Components
   - How to write Dockerfile for applications & Databases
   - How to write multi-stage Dockerfile
   - How to build Dockerfile
5. Storage
   - What are volumes
   - Types of volumes
   - Volume commands
   - Where to use volumes
   - How to Mount volumes to containers
6. Networks
   - What are Networks
   - Types of Types of Networks
   - Network commands
   - Where to use Networks
   - How to attach/detach networks to containers
7. Docker Compose
   - What is Docker Compose
   - Why Compose is needed
   - Docker Compose File Structure
   - Core Concepts
   - Key Commands
8. Docker Swarm
   - What is Docker Swarm
   - Swarm Architecture
   - Key Features
   - Swarm Initialization
   - Node Management
   - Services in Swarm
   - Scaling
   - Updating Services
9. Docker Stack
   - What is Docker Stack
   - Stack Architecture
   - Deploy Stack
   - View Stack
   - 10. Integrations
   - Integrating Docker With Jenkins to automate the deployment

KUBERNETES

1. Why Kubernetes Exists
   - Problems with running containers manually
   - Scaling issues in Docker
   - Need for orchestration
2. Introduction to Kubernetes
   - What is Kubernetes?
   - Google origin (Borg system concept)
   - Why DevOps uses K8s
3. Kubernetes Architecture
   - etcd
   - API server request flow
   - Scheduler internals
   - Controller manager
   - Kubelet
   - Kube-proxy
   - Pod lifecycle inside cluster

- 4. Kubernetes Objects
  - Pods
  - Services
  - Deployments
- 5. Basic Kubectl Commands
  - kubectl get pods
  - kubectl get nodes
  - kubectl describe pod
  - kubectl logs
  - kubectl apply -f
  - kubectl delete
- 6. YAML in Kubernetes
  - Why YAML is used
  - Structure of manifest files
- 7. Pods
- 8. Labels & Selector
- 9. Services
- 10. ReplicationController
- 11. ReplicaSets
- 12. Deployments
- 13. HPA
- 14. StatefulSet
- 15. DaemonSet
- 16. ConfigMaps
- 17. Secrets
- 18. Storage
  - Volumes
  - PV
  - PVC
  - 19. Namespaces
  - Default
  - Kube-system
  - Kube-public
  - Kube-nodelease
  - Custom Namespaces
- 20. Ingress
- 21. RBAC
  - Creating Users in Cluster
  - Role
  - Role Binding
  - Cluster Role
  - Cluster Role Binding
  - User Accounts
  - Service Accounts
- 22. Resource Quota
  - Requests
  - Limits
- 23. Helm
  - Intro
  - Installation of Helm
  - Downloading the helm repos
  - Install helm charts
  - Creating Own helm charts
  - Hosting Helm Charts
- 24. GITOPS (ArgoCD)
  - GitOps workflow model
  - ArgoCD sync strategies
  - rollback via Git
  - drift detection
- 25. Network Policies
  - Introduction
  - CNI used as calico

- Policy Types (ingress & egress)
- PodSelector & Namespace Selector

26. Affinities
- Pod Affinity
- Pod Anti-Affinity
- Node Affinity

27. Node selector

28. Taint & Tolerence

29. PodDisruptionBudget

30. Deployment Strategies
- canary
- blue-green
- recreate
- rolling updates

31. Probes
- Introduction
- LivenessProbe
- Readiness Probe
- Startup Probe

32. Jobs
- Introduction
- Jobs
- Parallelism
- CronJobs

33. Observability Stack
- Prometheus
- Grafana
- Loki
- Promptail
- Metrics vs Logs vs Traces
- Alerting (Alertmanager)

34. Backup & Disaster Recovery
- etcd backup & restore
- Velero (cluster backup tool)
- DR strategies

35. Cluster Upgrades & Lifecycle Management
- Kubernetes version upgrade strategy
- node draining
- rolling cluster upgrade
- backward compatibility issues

36. Troubleshooting Section
- CrashLoopBackOff
- ImagePullBackOff
- Pending pods
- Node not ready
- kubectl debug
- Events troubleshooting

TERRAFORM

1. Why Infrastructure Automation?
- What is infrastructure?
- Manual infrastructure vs automated infrastructure
- Problems in manual provisioning

2. Introduction to Infrastructure as Code (IaC)
- What is IaC?
- Declarative vs Imperative approach
- Benefits of IaC

3. What is Terraform?
   - Terraform overview
   - HashiCorp Terraform
   - Why Terraform is popular in DevOps
   - Cloud-agnostic tool
4. Terraform Architecture
   - Terraform CLI
   - Providers
   - Resources
   - State file
5. Installation & Setup
   - Install Terraform
   - Verify installation
6. Creating a resource using Terraform
   - Simple AWS resource (EC2 / S3)
7. Terraform Core Commands
   - terraform init
   - terraform plan
   - terraform apply
   - terraform destroy
8. Providers & Resources
   - What is a provider?
   - What is a resource?
   - AWS provider example
9. Variables
   - Input variables
   - Default values
   - tfvar files
   - CLI variables
10. Outputs
    - Display results after deployment
11. State File
    - What is terraform.tfstate?
    - Why state is important
    - Local vs remote state
12. Remote State
    - S3 backend
    - State locking (DynamoDB)
13. Terraform Lifecycle
    - create
    - update
    - delete
    - drift detection
14. Modules
    - What is a module?
    - Reusability concept
    - Root module vs child module
15. Workspaces
    - Multiple environments
    - Commands on workspaces
16. Data Sources
    - Fetch existing infrastructure
17. Provisioners
    - file provisioner
    - remote-exec
    - local-exec
18. Dependency Management
    - implicit dependencies
    - explicit dependencies (depends_on)

19. Terraform with AWS
- VPC creation
- EC2 deployment
- S3
- ASG
- Security groups
- Load balancer setup

20. Terraform + CI/CD Integration
- Jenkins + Terraform

21. Terraform State Deep Dive
- State locking
- State corruption handling
- State recovery
- Backend strategies

22. Security in Terraform
- secrets management
- sensitive variables
- IAM least privilege
- tfsec scanning

23. Terraform Drift Detection
- infrastructure drift
- remediation strategies

24. Import Existing Infrastructure
- terraform import
- migrating manual infra → IaC

SECURITY TOOLS
- Sonarqube
- Open Source Web Application Secure Project
- Trivy
- Istio
- HCP
- TF SEC

MCP WITH DEVOPS
- Jenkins With MCP
- Docker With MCP
- K8s with MCP
- Terraform with MCP

SELF PACED MODULE
- Ansible
- New relic

DELIVERABLES

1.FROM SCRATCH TO MASTER LEVEL TRAINING

2.RESUME BUILDING

3.DAILY ASSIGNMENTS

4.LINKEDIN PROFILE BUILDING

5.INTERVIEW GUIDANCE

6.COURSE COMPLETION CERTIFICATE

7.Q&A SESSIONS

8.PLACEMENT UPDATES

9.ON-DEMAND VIDEO COURSE CONTENT

10.DOWNLOADABLE RESOURCES

TOP CITIES FOR DEVOPS & CLOUD DEMAND

- BANGALORE (BENGALURU)
- GURGAON (GURUGRAM)
- HYDERABAD
- PUNE
- MUMBAI
- CHENNAI
- NOIDA

KEY SECTORS DRIVING DEMAND FOR DEVOPS PROFESSIONALS

- TECHNOLOGY AND SOFTWARE DEVELOPMENT
- E-COMMERCE AND RETAIL
- FINANCE AND FINTECH
- HEALTHCARE AND PHARMA
- EDUCATION AND EDTECH
- DATA SCIENCE AND ANALYTICS
- ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING
- MEDIA AND ENTERTAINMENT
- STARTUPS AND TECH INCUBATORS

TOP COMPANIES HIRING



















ALL THE BEST

